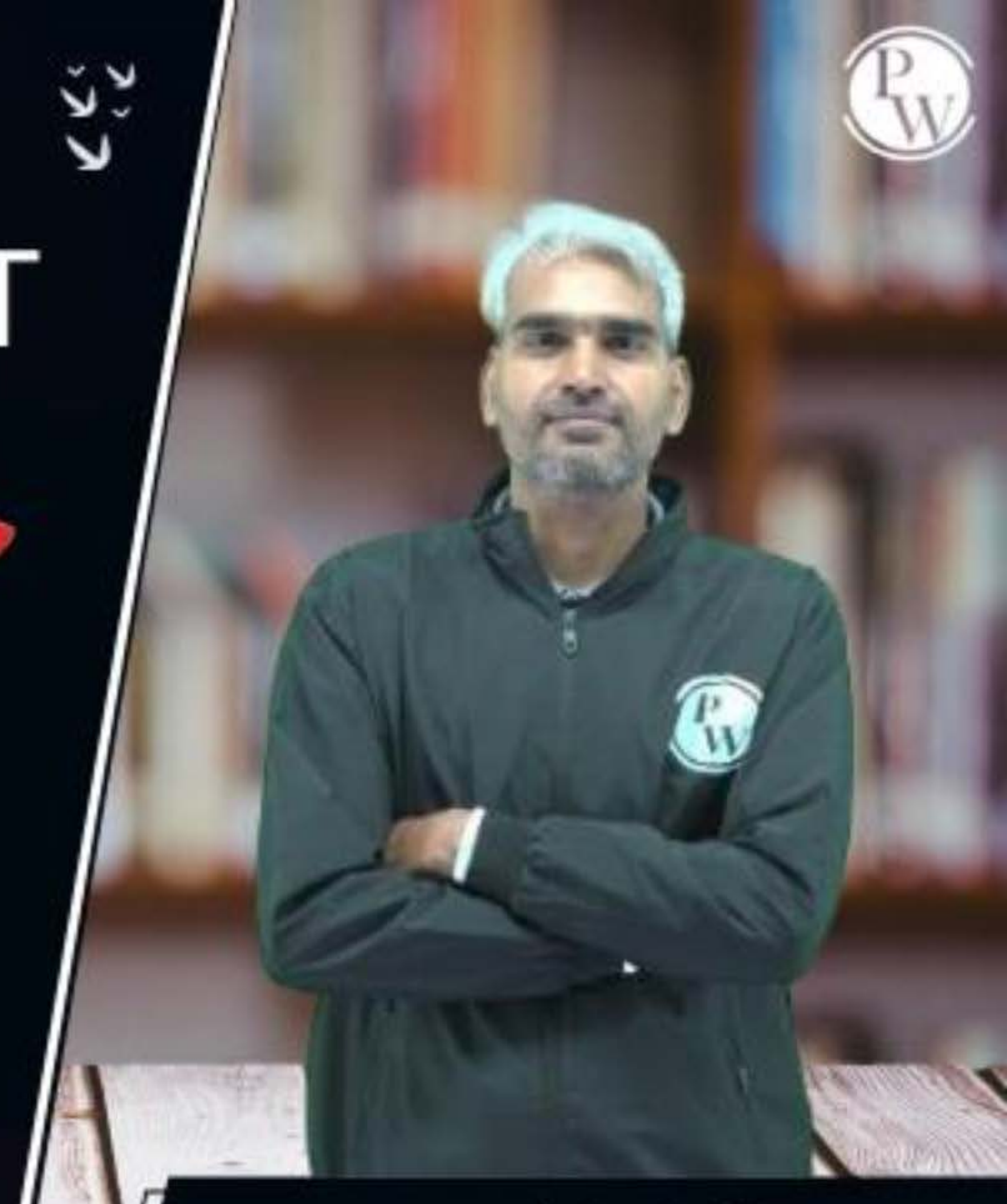
Computer Science & IT

Database Management System

Query Languages

Lecture No. 06



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Recap of Previous Lecture

Topic Practice questions on R.A.

Topic SQL

Topic SQL clauses

Slide

Topics to be Covered

- **Topic** Aggregate functions
- **Topic** SQL commands
- **Topic** SQL clauses

Slide

Prerequisite :- Relational Algebra

Topic : SQL

→ Structured Query language

$\underbrace{\text{Structured Query language}}$ → It is a $\underbrace{\text{non-procedural}}$ query language

∴ We must understand the "Syntax" of SQL

Topic : Commands in SQL

Commands in SQL

- Create, Drop, Alter — Data Definition Language (DDL)
- Insert, Delete, Update, Truncate — Data Manipulation Language (DML)
- Grant, Revoke — Data Control Language (DCL)
- Commit, Rollback/Abort — Transaction Control Language (TCL)
- Select — Data Query Language (DQL)

Topic : SQL clauses

* Some basic SQL Clauses are :-

1. From
2. Where
3. Group By
4. Having
5. Order By

Topic : SQL clauses

it is optional.

If we do not use "distinct" then O/P of SQL query may contain duplicate tuples

$\pi_{A_1, A_2, \ldots A_k} \equiv$ ① Select [distinct] $\underbrace{A_1, A_2, \ldots A_k}$

List of attributes required in O/P

$R_1 \times R_2 \times \ldots \times R_m \equiv$ ② From $\underbrace{R_1, R_2, \ldots, R_m}$

List of relations required for the execution of query

$\sigma_{(\text{Cond}^n \text{ to select tuples})} \equiv$ ③ Where (Cond^n to select tuples)

No Equivalent R.A. Expression $\equiv$ ④ Group by (Attibutes)

⑤ Having (Cond^n)

Cond^n specified with "having" clause will be applied on each group, provided "Group by" clause is present

⑥ Order by (Attribute) DESC/ASC

Select $\not\equiv$ Projection (π)
(in SQL) (in Relational Algebra)

Select distinct $\equiv$ Projection (π)
(in SQL) (in Relational Algebra)

Topic : Aggregate functions

H.W.

- Select Count (*)
  from STUDENT = ?

- Select Count (Marks)
  from STUDENT = ?

- Selct Count (distinct Marks)
  from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

HW

- Select Sum(Marks)
  from STUDENT = ?

- Select Sum(distinct Marks)
  from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

H.W.

- Select Avg(Marks)
  From STUDENT = ?

- Select Avg (distinct Marks)
  From STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

H.W.

- Select Max(Marks)
  from STUDENT = ?

- Select Max (distinct Marks)
  from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

H.W.:

- Select Min (Marks) from STUDENT = ?

- Select Min (distinct Marks) from STUDENT = ?

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

* There are five aggregate functions available in SQL

1. Count( )
2. Sum( )
3. Avg( )
4. Max( )
5. Min( )

Topic : Aggregate functions

Every-thing

- Count(*) - It returns the total number of tuples in the relation. Count(*) will also count the empty tuples, i.e., the tuples in which all values all NULL.

- Count(Attribute) - It will count the total number of non NULL values in the column corresponding to the attribute specified with count function.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : NOTE

- NULL is a non-zero unknown value.
- No two NULL are equal. {i.e, we can not compare NULL with NULL}
- NULL values are always discarded by aggregate functions except for count(*).
- Arithmetic operation with NULL will always produce NULL

  eg. NULL + 50 = NULL

Topic : Aggregate functions

* Select Count(*)
  from STUDENT = 7

* Select Count (Sid)
  from STUDENT = 6

* Select Count (Marks)
  from STUDENT = 5

* Select Count (distinct Sid)
  from STUDENT = 6

* Select Count (distinct Marks)
  from STUDENT = 4

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

- Sum(Attribute) - It will return the summation of all non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

- Sum(distinct Attribute) - It will return the summation of all distinct non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

* Select Sum (Marks) from STUDENT = 250

* Select Sum (distinct Marks) from STUDENT = 190

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60 ✗	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

- **Avg(Attribute)** - It will return the average of all non NULL values with respect to specified attribute.

$$\text{Avg(Attribute)} = \frac{\text{SUM(Attribute)}}{\text{Count(Attribute)}}$$

- **Avg(distinct Attribute)** - It will return average of all distinct non NULL values with respect to specified attribute.

$$\text{Avg(distinct Attribute)} = \frac{\text{SUM(distinct Attribute)}}{\text{Count(distinct Attribute)}}$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

* Select Avg (Marks)
  from STUDENT =

$$= \frac{SUM(Marks)}{Count(Marks)} = \frac{250}{5} = 50$$

* Select Avg (distinct Marks)
  from STUDENT =

$$= \frac{SUM(distinct\ Marks)}{Count(distinct\ Marks)} = \frac{190}{4} = 47.5$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

- Max(Attribute) - It will return the maximum of all the values in the column corresponding to the specified attribute.

- Min(Attribute) - It will return the minimum of all the values in the column corresponding to the specified attribute.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Aggregate functions

- Select Max (Marks)
  from STUDENT = 70

- Select Min (Marks)
  from STUDENT = 20

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL

Topic : Commands in SQL

- CREATE TABLE - creates a new table
- ALTER TABLE - modifies a table ⟹ { used to ADD or Delete the Columns in table }
- DROP TABLE - deletes a table
  ↳ It will delete the complete table.

- INSERT - inserts new data (tuples) into a database
- DELETE - deletes data from a database
- UPDATE - updates data in a database

- SELECT - extracts data from a database

Topic : Commands in SQL

- CREATE TABLE - creates a new table

$$\underbrace{\text{CREATE TABLE}}_{\text{Command}}\ \underbrace{\text{Student}}_{\text{name of table to be created}}\ ($$

```
CREATE TABLE Student (
        Sid varchar(25),
        Sname varchar(25),
        Marks int,
                    );
```

attributes

data type

Create table

new_table_name

(Attribute$_1$ data_type Constraint,

Attribute$_2$ data_type Constraint,

⋮ ⋮ ⋮

);

Topic : Commands in SQL

- CREATE TABLE - creates a new table

```
CREATE TABLE Student (
        Sid varchar(25),
        Sname varchar(25),
        Marks int,
                );
```

Student

Sid	Sname	Marks

Topic : Commands in SQL

- ALTER TABLE - modifies a table

Add new Column
(or)
delete existing Column

ALTER TABLE *Student*
ADD $\underbrace{\textit{Branch}}_{\text{name of new attribute}}$ $\underbrace{\textit{varchar(25)}}_{\text{data\_type}}$;

we can also define Contraint.

Student

Sid	Sname	Marks

Branch

- To "ADD" a new Column we use ADD
- To "delete" an Existing Column we use DROP

Topic : Commands in SQL

- ALTER TABLE - modifies a table

```
ALTER TABLE Student
ADD Branch varchar(25);
```

Student

Sid	Sname	Marks	Branch

If we write

Alter table Student
Drop Sname;

used to delete

$\Rightarrow$ As a result of this query, Column "Sname" will get deleted

Topic : Commands in SQL

- DROP TABLE - deletes a table, { Delete the Complete table along with its structure }

DROP TABLE *Student*; ⇒ As a result of this query Student table will be deleted from the database

$\underbrace{\text{DROP TABLE}}_{\text{Command}}$ $\underbrace{\textit{Student}}_{\text{table\_name}}$;

- TRUNCATE TABLE - deletes complete data from the table without deleting the structure of the table

TRUNCATE TABLE Student;

$\underbrace{\text{TRUNCATE TABLE}}_{\text{Command}}$ $\underbrace{\text{Student}}_{\text{table\_name}}$;

Topic : Commands in SQL

- **INSERT** - inserts new data(row/rows) into a database table.

Command table_name

- INSERT INTO *Student*

VALUES (*S1*, *A*, *35*, CS);

Value w.r.t. 1st attribute

Value w.r.t 2nd Attribute

and so on.

Student

Sid	Sname	Marks	Branch

Insert Into table_name
(Values (Att1, Att2, ...),
Values (Att1, Att2, ...),
⋮
);

We can insert multiple tuples using a single SQL query.

Topic : Commands in SQL

- INSERT - inserts new data(row/rows) into a database table.

- INSERT INTO *Student* VALUES (*S1*, *A*, *35*, CS);

Student

Sid	Sname	Marks	Branch
S1	A	35	CS

Topic : Commands in SQL

- DELETE - deletes data(row/rows) from a database table

eg. Delete from Student
Where (Marks = 35)

← All the tuples in which Marks are "35" will get deleted

Topic : Commands in SQL

- UPDATE - updates data in a database table

Comman table_name

UPDATE *Student*

SET *Marks = Marks + 5;*

Part of Update Command

Student

Sid	Sname	Marks	Branch
S1	A	35	CS

Update table_name

Set

Updation required

- UPDATE - updates data in a database

```
UPDATE Student
SET Marks = Marks + 5;
```

Student

Sid	Sname	Marks	Branch
S1	A	35+5=40	CS

Topic : SQL clauses

- **FROM** - The FROM clause in SQL is used to select the database tables.
- **WHERE** - The WHERE clause in SQL is used to select the tuples from the database table based on conditions specified with WHERE clause. {If where clause is not present, then all the tuples will be selected}
- **GROUP BY** - GROUP BY clause is used to group the result of WHERE clause based on attributes specified with GROUP BY clause.
  {If where clause is not present then group by clause will be applied on all tuples}
  {If Group By clause is not present then all tuples together will be treated as a single group}
- **HAVING** - HAVING clause can be used with or without GROUP BY clause. If group by clause is present, then condition specified with having clause will be used to select the groups that satisfy the specified condition. {Having Cond^n is applied on each group}
- **ORDER BY** - The ORDER BY clause in SQL is used for sorting the records of the database based on attributes specified with ORDER BY clause.

Topic : Order of execution

Order of Execution:-

1. From
2. Where
3. Group By
4. Having
5. Select

5' distinct

6. Order BY

Topic : SQL clauses

FROM:- From clause is used to select the tables from the database.

Everything {i.e; Select all attributes}

Select *

From Student

o/p of this query will be Complete Student table.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

✓ FROM:- From clause is used to select the tables from the database.

Select Sid
from Student

Select distinct Sid
from Student

⇒ o/p :-

Sid
S_1
S_2
S_3
S_4
S_5
S_6

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

✓ FROM:- From clause is used to select the tables from the database.

Select Sname ⇒ o/p
From Student

Sname
A
A
B
A
C
C

Select distinct Sname
From Student
↳ o/p =

Sname
A
B
C

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

If where clause is not present all tuple will be selected

WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

* Retrieve tuple from Student table in which Marks are more than 40

③ Select *
① From Student
② Where Marks > 40

o/p=

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

If where clause is not present all tuple will be selected

* Retrieve Sids of students from Student table where marks are more than or equal to 20 and less than or equal to 40

Select Sid
from Student
where (Marks ≥ 20 AND Marks ≤ 40)

O/p =

Sid
S1
S2
S5

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

If where clause is not present all tuple will be selected

WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

* Retrieve Sids of students from Student table where marks are more than or equal to 20 and less than or equal to 40

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Select Sid
from Student
Where Marks Between 20 AND 40

Both are included with Between

O/p :

Sid
S1
S2
S5

Topic : SQL clauses

- GROUP BY:- GROUP BY clause is used to group the result of WHERE clause. {If Where clause is not present all the tuples of the table are grouped based on attributes specified}

Query: Retrieve names of all branches along with maximum marks in that branch.

H.W.:-

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

NOTE:-

1. We can not select any attribute in SELECT clause along with aggregate function if those attributes are not present in GROUP BY clause.

2. If aggregate function is used along with GROUP BY clause, then aggregate function is applied on each group.

Topic : SQL clauses

GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

Query: Retrieve names of all branches along with maximum marks in that branch.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

HAVING:- HAVING condition is applied on each group.

Query: Retrieve branch names with average marks more than or equal to 40.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : SQL clauses

NOTE:-

1. WHERE condition is applied on each tuple whereas HAVING condition is applied on each group.
2. We can use HAVING condition without GROUP BY clause as well, but in that case all the tuples together are treated as a single group.

Topic : SQL clauses

ORDER BY:- This clause is used to sort the result in ascending or descending order based on values of attribute specified with ORDER BY clause.

By default order is ascending order.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Topic : Order of execution

Order of Execution:-

1. From
2. Where
3. Group By
4. Having
5. Select
6. Order BY

2 mins Summary

- **Topic** Aggregate functions
- **Topic** SQL commands
- **Topic** SQL clauses

Slide

THANK - YOU

Slide